

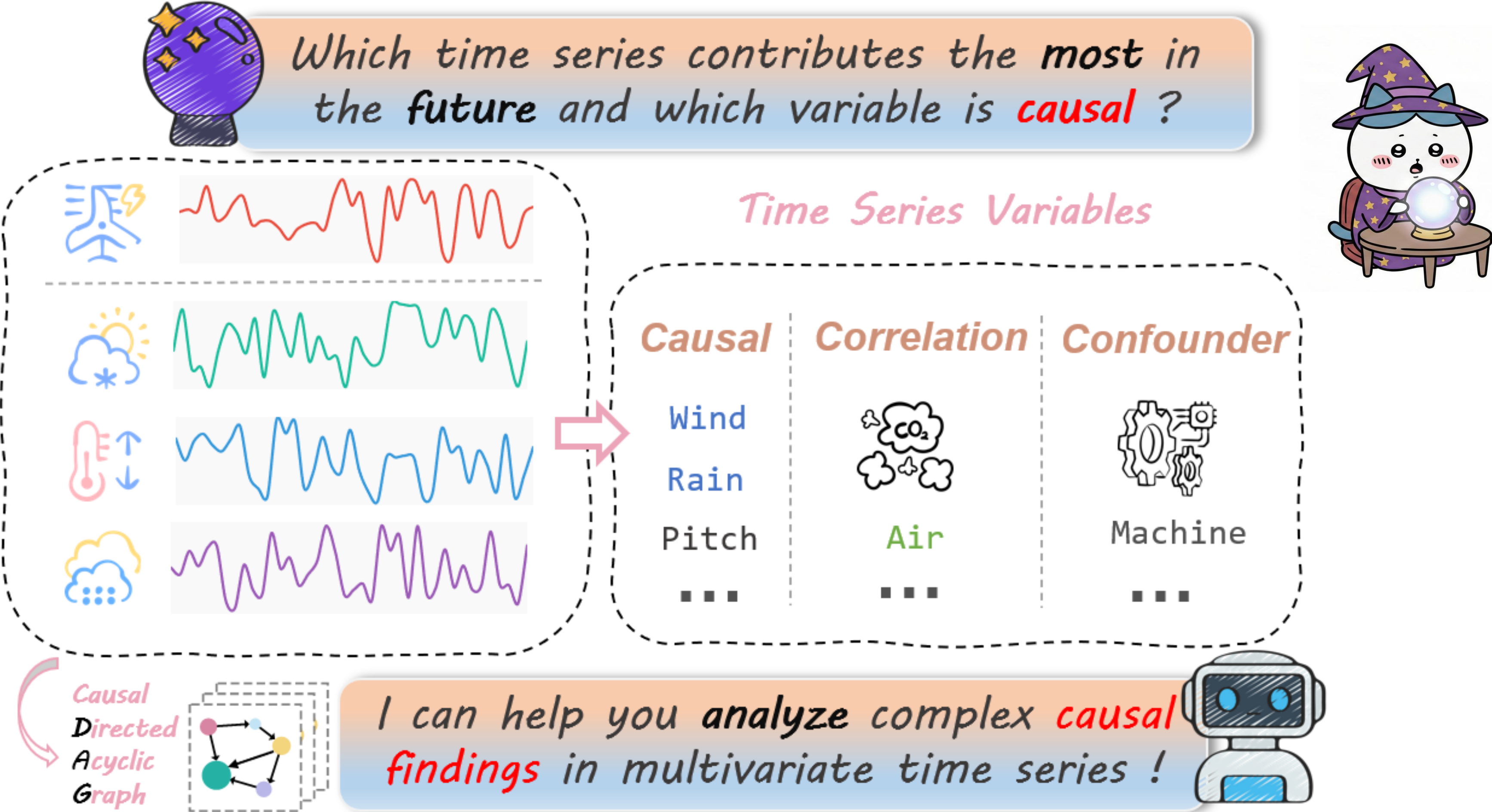
Augur: Modeling Covariate Causal Associations in Time Series via Large Language Models

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Introduction

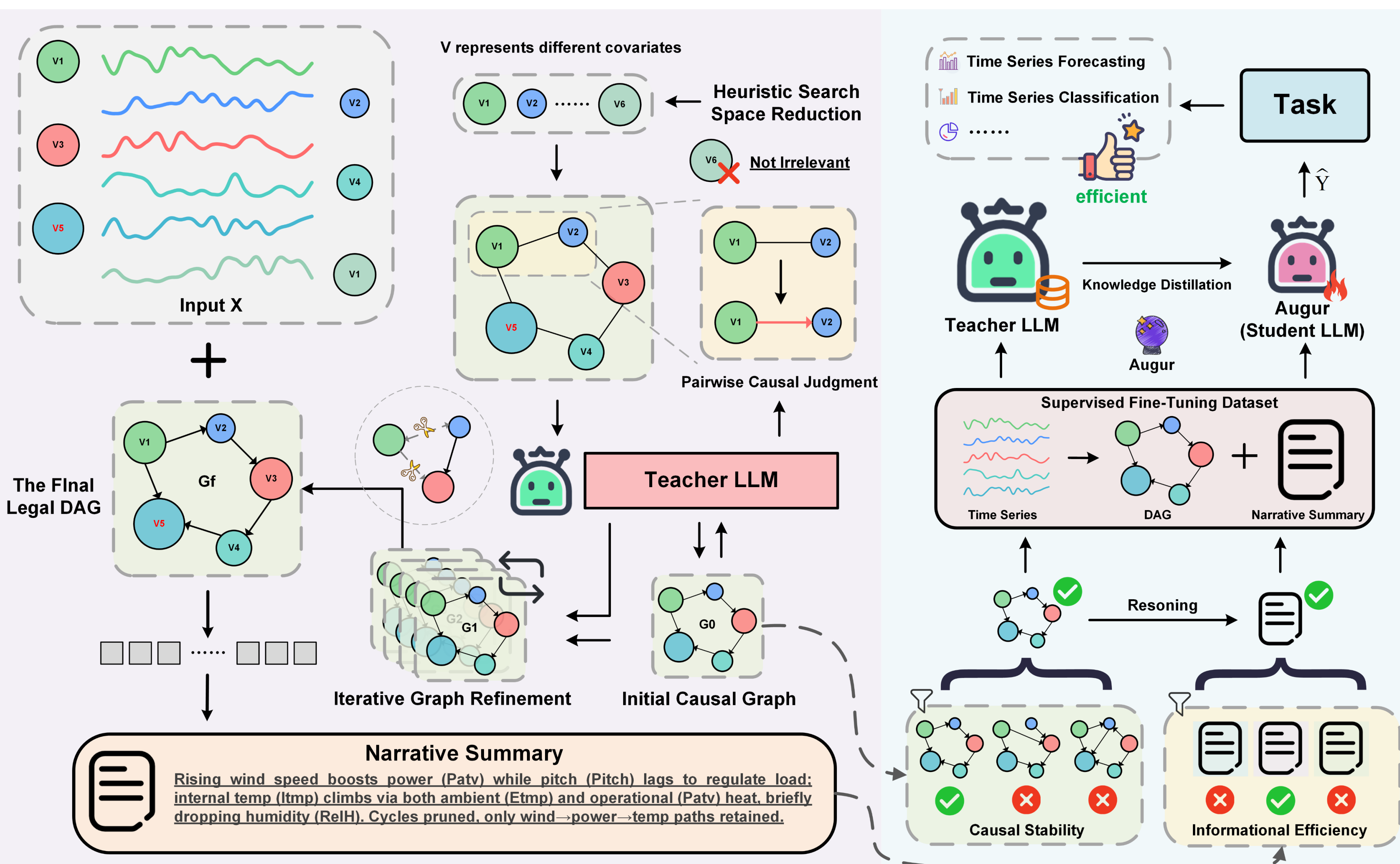


Previous Models: Rely on marginalized LLMs and ST-models without structural causal knowledge Interpretable ✗ Reasoning ✗

Casual LLM: teacher-student architecture to model directed causal associations among covariates Interpretable ○ Reasoning ○

Method

Modeling Covariate Causal Associations in Spatiotemporal via LLM



The inherent opacity of current state-of-the-art forecasting models, such as Spatiotemporal Graph Neural Networks (STGNNs). While these models achieve high numerical accuracy in predicting "what" will happen, they fail to explain "why" specific events occur. This lack of interpretability creates a critical trust bottleneck in high-stakes domains like **meteorology** and **urban planning**, where decision-makers require actionable, **causal insights** rather than "black-box" predictions.

Experiment

Prediction Performance Study

Model	Air		Power		Traffic		Finance	
	F1-Score	AUROC	F1-Score	AUROC	F1-Score	AUROC	F1-Score	AUROC
Time-VLM	0.845	0.918	0.795	0.852	0.692	0.775	0.672	0.748
Time-LLM	0.826	0.907	0.744	0.796	0.617	0.686	0.609	0.708
GPT4MTS	0.803	0.864	0.716	0.777	0.562	0.621	0.581	0.652
Moirai	0.876	0.928	0.797	0.858	0.706	0.787	0.676	0.767
Chronos	0.839	0.921	0.789	0.849	0.698	0.769	0.665	0.744
Time-MoE	0.891	0.941	0.818	0.879	0.718	0.803	0.681	0.762
Informer	0.846	0.903	0.764	0.844	0.676	0.748	0.646	0.717
Autoformer	0.803	0.908	0.757	0.836	0.684	0.756	0.653	0.724
FEDFormer	0.859	0.912	0.781	0.841	0.688	0.761	0.626	0.707
Crossformer	0.844	0.897	0.767	0.840	0.681	0.733	0.658	0.711
DLinear	0.745	0.841	0.642	0.748	0.516	0.596	0.523	0.615
iTransformer	0.878	0.929	0.793	0.864	0.713	0.764	0.684	0.755
PatchTST	0.885	0.936	0.823	0.881	0.724	0.803	0.696	0.748
LightTS	0.761	0.858	0.674	0.761	0.523	0.595	0.551	0.622
TimesNet	0.864	0.919	0.759	0.851	0.703	0.751	0.667	0.736
SparseTSF	0.640	0.793	0.628	0.755	0.531	0.612	0.502	0.531
PatchMixer	0.805	0.869	0.702	0.768	0.542	0.613	0.568	0.659
CycleNet	0.811	0.874	0.708	0.773	0.538	0.609	0.524	0.635
TimeMixer	0.889	0.939	0.813	0.884	0.735	0.806	0.692	0.763
TimeXer	0.873	0.924	0.795	0.855	0.706	0.785	0.672	0.744
Augur	0.928	0.958	0.849	0.909	0.751	0.825	0.705	0.783

Ablation Study

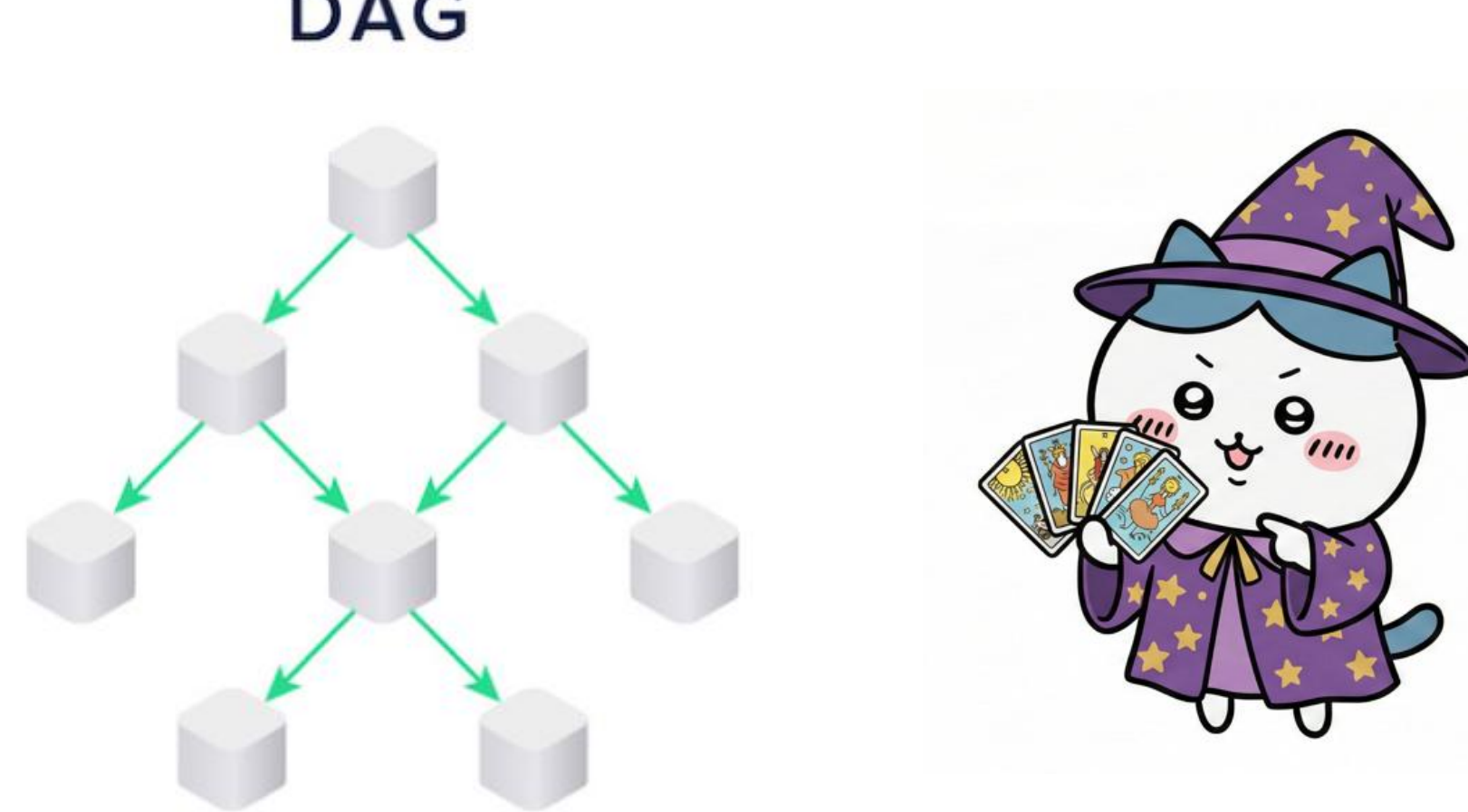
Variant	Time (x)	Prune	Judge	Refine	F1	AUC
w/o Prune	5.3	✗	✓	✓	0.81	0.88
w/o Judge	1.7	✓	✗	✓	0.76	0.84
w/o Refine	0.6	✓	✓	✗	0.79	0.87
Augur	1.0	✓	✓	✓	0.85	0.91

Human and Casual Evaluation

Dataset	Method	Evaluation Metrics				Method	Precision	Recall	F1-Score	SHD ↓
		EoU	Ins.	Corr.	Avg					
Power	GPT-4o	4.3	3.9	4.2	4.1	PC	0.15	0.10	0.12	11.8
	Qwen3	4.6	4.4	3.7	4.2	GES	0.15	0.26	0.19	18.7
	MI	0.09	0.95	0.17	76.1	Granger	0.15	0.54	0.24	26.6
	Augur	4.7	5.2	4.9	4.9	PCMC1	0.16	0.38	0.22	21.2
Air	GPT-4o	4.5	4.0	4.3	4.3	Augur	0.41	0.31	0.34	9.3
	Qwen3	5.1	4.5	3.8	4.5					
	Augur	4.9	5.5	5.0	5.1					

Contributions

- Moves beyond mere numerical correlations to identify underlying **physical drivers** and directed **causal associations** within complex dynamic systems
- Proposes the Casual-LLM **Teacher-Student** architecture to achieve advanced logical reasoning



! Looking Forward 27 Fall PhD. or RA position (Now Junior in DS)
💡 Interest: LLM Agent Reasoning, AI4Earth, Spatiotemporal Model

